

CSE221: Algorithms

Lab-08

Fall 2025

Assignment-08 Links:

codeforces Invitation Link:

<https://codeforces.com/contestInvitation/176bcdbfda4f5b6822688cde184e6f4b82826eda>

Alternative Link: <https://codeforces.com/contests/659533>

[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline: Jan 4, 2026 2:00 PM

No submission can be considered after the deadline ends.

Important Notes:

There is a change in the lab plan as follows:

- **Quiz-05 Date:** Dec 27, 2025, Dec 28, 2025 (as per your routine)
- **Quiz-05 Syllabus:** Remains the same
 - Assignment-06 and Assignment-07
 - One quiz covering both topics
 - Take Preparation for both assignment problems and outside assignment problems

Learning Outcome:

By the end of this lab, students will be able to identify, model, and solve optimization problems using greedy algorithm techniques, particularly in graph-based and scheduling scenarios. They will understand the greedy-choice property and how locally optimal decisions can lead to globally optimal solutions in problems such as Minimum

Spanning Tree (MST) construction and interval task scheduling. Students will also develop an understanding of Disjoint Set Union (DSU) as the underlying algorithmic mechanism used to maintain connectivity during MST construction. Overall, the lab aims to strengthen students' ability to recognize greedy problem structures, reason about correctness, and design efficient solutions for real-world optimization problems.